CONVOLUTION ALGEBRAS OF DOUBLE GROUPOIDS AND STRICT 2-GROUPS

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Abstract. Double groupoids are a type of higher groupoid structure that can arise when one has two distinct groupoid products on the same set of arrows. A particularly important example of such structures is the irrational torus and, more generally, strict 2-groups. Groupoid structures give rise to convolution operations on the space of arrows. Therefore, a double groupoid comes equipped with two product operations on the space of functions. In this article we investigate in what sense these two convolution operations are compatible. We use the representation theory of compact Lie groups to get insight into a certain class of 2-groups.

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1. Introduction

Given a Lie groupoid $\mathcal{G} \rightrightarrows M$ and a Haar system on $\mathcal{G}$ one can associate a C^*-algebra $C^*(\mathcal{G})$. This relationship is the fundamental link between Lie groupoids and noncommutative geometry. The algebra $C^*(\mathcal{G})$ can, in some sense, be thought of as the (noncommutative) algebra of functions on the differentiable stack $[M/\mathcal{G}]$. This point of view is justified by a theorem of Muly, Renault, and Williams [7] (Theorem 2.8) which says that if $\mathcal{G}$ and $\mathcal{H}$ are Morita equivalent groupoids (i.e. they represent the same stack), then the associated C^* algebras $C^*(\mathcal{G})$ and $C^*(\mathcal{H})$ are strongly Morita equivalent. In particular, if $\mathcal{G}$ is Morita equivalent to a manifold then $C^*(\mathcal{G})$ is strongly Morita equivalent to the algebra of functions on the smooth quotient space $M/\mathcal{G}$.

The objective of this article is to try to better understand C^* algebras of some higher structures (namely double groupoids). In principal these algebras should be a model for a type of "noncommutative groupoid."

To better explain what we mean by this, let us consider our model example: The noncommutative torus is a noncommutative space that arises from the convolution algebra of the action groupoid $\mathbb{Z} \ltimes S^1$ where $\mathbb{Z}$ acts on S^1 by irrational rotations. The groupoid composition gives rise to a convolution product on the vector space of compactly supported functions $C_c^\infty(\mathbb{Z} \ltimes S^1)$. As we mentioned earlier, we think of this algebra as a model for the algebra of "functions" on the space $S^1/\mathbb{Z}$. However, we should note that $S^1/\mathbb{Z}$, while not smooth, is a perfectly well-defined group. Since $C_c^\infty(\mathbb{Z} \ltimes S^1)$ represents the functions on this singular quotient, we should, in principle, expect it to inherit two algebra structures. One of the algebra structures should be analogous to the product arising from pointwise multiplication while the other should be analogous to the *group* convolution product.

Now observe that $\mathbb{Z} \ltimes S^1$ actually has two natural algebra structures. One, denoted $\circ$, arises from treating $\mathbb{Z} \ltimes S^1$ as an action groupoid while the other, denoted $\bullet$, arises from treating $\mathbb{Z} \ltimes S^1$ as a Cartesian product of groups. These two product operations are compatible in the sense that:

$$(a \circ b) \bullet (c \circ d) = (a \bullet c) \circ (b \bullet d) \tag{1.1}$$

for all $a, b, c, d \in \mathbb{Z} \ltimes S^1$, suitably composable. The operations $\circ$ and $\bullet$ give rise to two different convolution operations on $C_c^\infty(\mathbb{Z} \ltimes S^1)$ which we denote $\overset{\circ}{*}$ and $\overset{\bullet}{*}$. One of the main aims of this article is to consider the question "In what sense are $\overset{\bullet}{*}$ and $\overset{\circ}{*}$ compatible?"

The general context for this question is that of a double groupoid. A double groupoid is a groupoid object in the category of groupoids. From the point of view of stacks, such an object can be thought of as a (strict) groupoid in the category of stacks. From the noncommutative geometry point of view, the convolution algebras of a double groupoid should be a type of "noncommutative groupoid."

The primary feature of a double groupoid $\mathcal{G}$ is that it has two product operations $\bullet$ and $\circ$ and these two product operations are *compatible* in the sense that they satisfy Equation 1.1 whenever both sides of the equation are well-defined. If we choose Haar systems, then we get two convolution operations $\overset{\circ}{*}$ and $\overset{\bullet}{*}$ on $C_c^\infty(\mathcal{G})$. Now let us break the symmetry of the situation by considering $\overset{\circ}{*}$ to be the algebra of functions on the noncommutative space defined by $\circ$. Then, intuitively, $\overset{\bullet}{*}$ should correspond to "convolution."

When discussing the compatibility of $\overset{\circ}{*}$ and $\overset{\bullet}{*}$, the naive guess would be to assume that they satisfy a version of Equation 1.1. However, the situation is not so simple. There is well-known lemma of Eckmann and Hilton which provides us with some hints as to why one should be careful:

Lemma 1.1 (Eckmann-Hilton[4])**.** *Suppose a set A is equipped with globally defined, unital, binary operations $\circ$ and $\bullet$ and assume that these two products satisfy Equation 1.1. Then $\circ$ is associative, commutative, and $\circ = \bullet$.*

At first glance, this lemma seems to suggest that there are no interesting double groupoids. However, the key point is that the binary operations in the Eckmann-Hilton lemma must be globally defined. Furthermore, a problem arises once we pass to the convolution algebras $\overset{\bullet}{*}$ and $\overset{\circ}{*}$ which are both globally defined on $C_c^\infty(\mathcal{G})$. Hence, the lemma of Eckmann and Hilton suggests that the naive notion of compatibility between $\overset{\circ}{*}$ and $\overset{\bullet}{*}$ does not hold.[1].

There two main aims for this paper. One is to establish some basic facts and definitions about the general case. However, as we will see, a comprehensive study of the compatibility properties for the general case is likely very complicated. After discussing the general case, we will study three main examples. The goal with our examples is to find formulas resembling Equation 1.1 that *are* true for the two convolution products. Our three main examples are (1) the irrational torus, (2) compact Lie groups (thought of as a trivial double groupoid) and (3) "compact singular Lie groups." Our strategy for all three of these cases will be to take advantage of the Fourier theory and representation theory of compact groups. The last case is the most general and encompasses the previous two cases.

Suppose $\mathcal{G} = K \ltimes G$ where K is a (possibly non-closed) discrete normal subgroup of a compact Lie group G. We let $\circ$ denote the composition operation on $\mathcal{G}$ from the action groupoid structure and let $\bullet$ denote the group product. Such a $\mathcal{G}$ is what we call a *compact singular Lie group*. Now let $\overset{\circ}{*}$ and $\overset{\bullet}{*}$ be the associated convolution operations on $C_c^\infty(\mathcal{G})$.

In order to explain the sense in which $\overset{\bullet}{*}$ and $\overset{\circ}{*}$ are compatible we will take advantage of a natural basis for $C_c^\infty(K \ltimes G)$ that comes from Fourier theory. Let $\widehat{G}$ denote the set of (equivalence classes) of irreducible representations of G. For each $\pi \in \widehat{G}$ and $0 \leq i, j \leq \dim \pi$ we can associate a function $\pi_{ij} \in C^\infty(G, \mathbb{C})$ by taking the matrix coefficients of the representation. Now for $\kappa \in K$ let:

$$\pi_{ij}^\kappa \in C_c^\infty(K \ltimes G, \mathbb{C}) \qquad \pi_{ij}^\kappa(\ell, g) := \begin{cases} 0 & \kappa \neq \ell \\ \pi_{ij}(g) & \kappa = \ell \end{cases}$$

Theorem 1.2. *The set*

$$\mathcal{B} := \{\pi_{ij}^\kappa \ : \ \pi \in \widehat{G}, \kappa \in K, 0 \leq i, j \leq \dim(\pi)\}$$

generates a dense subalgebra of $C_c^\infty(K \ltimes G, \mathbb{C})$ Furthermore, for all:

$$\pi, \sigma \in \widehat{G}, \qquad \kappa_1, \kappa_2, \lambda_1, \lambda_2 \in K, \quad 0 \leq i, k \leq n, \quad 0 \leq a, c \leq m$$

[1]Since convolution algebras are not typically unital, we cannot directly apply the Eckmann-Hilton lemma here. However, convolution algebras are not that far from being unital and one could reasonably expect that the argument can be adapted to this setting.

we have that

$$\sum_j \sum_b (\pi_{ij}^{\kappa_1} \overset{\bullet}{*} \pi_{jk}^{\kappa_2}) \overset{\circ}{*} (\sigma_{ab}^{\lambda_1} \overset{\bullet}{*} \sigma_{bc}^{\lambda_2}) = \sum_j \sum_b (\pi_{ij}^{\kappa_1} \overset{\circ}{*} \sigma_{ab}^{\lambda_1}) \overset{\bullet}{*} (\pi_{jk}^{\kappa_2} \overset{\circ}{*} \sigma_{bc}^{\lambda_2}). \tag{1.2}$$

If the above theorem did not have sums, it would describe a set of 4-tuples where a compatibility condition of the form 1.1 holds. The stronger version of Equation 1.2 (i.e. without the sums) is false in general. However, it holds for some of the more tame examples such as the case where G is a torus.

The reason one might expect such a formula to be true is due to the existence of a discrete groupoid that functions as the "Fourier transform" of $\mathcal{G}$ (we explore this concept in Sections 8.1 and 9.1). It turns out that for discrete groupoids, compatibility on generators is governed by composability. The above equation, is a kind of statement about compatibility for "composable" generators. An interpretation of the above formula is to say that the failure of the compatibility law for "composable" generators vanishes "on average."

1.1. **Additional remarks.** To the best of our knowledge, there is not much by way of existing literature on convolution algebras for double groupoids. Sections 2-5 are largely dedicated to ensuring that this article can be a relatively self-contained introduction to the topic.

Sections 6-9 are dedicated to using the representation theory of compact Lie groups to compute some interesting examples. These calculations we do here lean heavily on the fact that the representation theory of compact Lie groups admits some very useful simplifications. This allows us to find a suitable basis for the convolution algebra where we can perform calculations. In the future it could be interesting to better understand the case of non-compact Lie groups where the representation theory can be significantly more subtle.

We should also make a remark on the topic of compatible Haar systems. The examples for which we perform computations have Haar systems that are sufficiently canonical that we do not seem to encounter many issues with ensuring that the Haar systems work well together. However, for more general cases, a notion of compatible Haar systems is surely necessary. A definition of a 2-Haar system appears in Amini [1] but there doesn't seem to be any compatibility condition.

The compatibility law (Equation 1.2) that we have proved relies quite a lot on specific facts about the structure of the convolution algebras involved. We suspect that there is a more general form which might lead to a notion of "compatible algebras." However, at this time, the correct notion of compatible algebras is not completely clear.

There are also alternative approaches one could take to investigate this topic. Rather than considering two algebra structures on the same set, one could instead attempt to study these structures in the form of bialgebras or Hopf algebras. Some work in this direction does exist (see for example Hopfish algebras [9][2]).

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2. Groupoids background

In this section we will review some basic facts about and establish our notation for Lie groupoids and related algebraic structures.

2.1. **Categories.** The most general kind of algebraic object we will examine is that of a (small) category. Let us give a definition so that we can establish some of our notation conventions and terminology for categories. Our conventions will reflect the fact that we will be considering categories as algebraic objects in the vein of a group or a monoid.

Definition 2.1. [Category] A (small) category $C = C_1 \rightrightarrows C_0$ is a pair of sets C_1 (the *arrows*) and C_0 (the *objects*) together with a pair of functions

$$s\colon C_1 \to C_0 \qquad t\colon C_1 \to C_0$$

called the *source* and *target*. Another function: a function

$$u\colon \mathcal{C}_0 \to \mathcal{C}_1 \qquad x \mapsto 1_x$$

called the *unit*. The set of composable arrows is defined to be:

$$C_1 \times_{s,t} C_1 := \{(a,b) \in C_1 \ : \ s(a) = t(b)\}$$

and we have a function:

$$m\colon C_1 \times_{s,t} C_1 \to C_1 \qquad (a,b) \mapsto a \circ b$$

called *multiplication* or *composition*. We require that these functions satisfy the following axioms:

- (Compatibility of source and target with multiplication)
$$\forall (a,b) \in \mathcal{C}_1 \times_{s,t} \mathcal{C}_1 \qquad t(a \circ b) = t(a) \quad \text{and } s(a \circ b) = s(b)$$
- (Associativity)
$$\forall (a,b,c) \in \mathcal{G} \times_{s,t} \mathcal{G} \times_{s,t} \mathcal{G} \qquad (a \circ b) \circ c = a \circ (b \circ c)$$
- (Compatibility of the unit with source and target)
$$\forall x \in \mathcal{C}_0 \qquad s(1_x) = t(1_x)$$
- (Left and right unit laws)
$$\forall a \in \mathcal{C}_1 \qquad 1_{t(a)} \circ a = a = a \circ 1_{s(a)}$$

If the set of objects $\mathcal{C}_0$ is a singleton, then we say that $\mathcal{C}$ is a *monoid*. If we remove the unit map and its associated axioms the resulting structure is called a *semi-category* and a semi-category with only one object is called a *semi-monoid* (Definition 2.1).

In general we will use the notation $A \rightrightarrows B$ to indicate that A is the arrows of a category with objects B. Let us now consider a few basic examples of categories that will play a role in our discussion.

Example 2.2. Let $\mathcal{C}_0 = \{*\}$ and let $\mathcal{C}_1 = \mathbb{N}$. If we take composition to be addition of natural numbers, then this constitutes a category. Since it only has one object, it is an example of a monoid.

Example 2.3 (Complex Euclidean representations)**.** Let G be a Lie group and let

$$\mathcal{C}_0 = \{\rho\colon G \to \mathrm{GL}(n,\mathbb{C}) \ : \ n \in \mathbb{N},\ \rho \text{ representation } \}$$

be the set of all representations of G on complex Euclidean spaces. Given an object $\rho \in \mathcal{C}_0$ let us write V_ρ to denote the underlying complex Euclidean space. Now let:

$$\mathcal{C}_1 := \{(\rho_2, L, \rho_1) \ : \rho_1, \rho_2 \in \mathcal{C}_0,\ L\colon V_{\rho_1} \to V_{\rho_2} \text{ linear intertwining operator } \}.$$

Here, linear intertwining operator means that L is a linear map and for all $g \in G$, $v \in V_{\rho_1}$ we have that:

$$L(\rho_1(g)v) = \rho_2(g)L(v)$$

It is not too difficult to see that this is an example of a category if we take the multiplication operation to be:

$$(\rho_3, L_2, \rho_2) \circ (\rho_2, L_1, \rho_1) = (\rho_3, L_2L_1, \rho_1).$$

2.2. **Groupoids.** Briefly, a groupoid is a category $\mathcal{G}$ where every arrow is invertible. In our notation conventions we will typically use $\mathcal{C}$ to denote a category and $\mathcal{G}$ to denote a groupoid.

Definition 2.4 (Groupoid)**.** A groupoid $\mathcal{G}$ is a category equipped with a function:

$$i\colon \mathcal{G}_1 \to \mathcal{G}_1 \qquad g \mapsto g^{-1}$$

which satisfies the following properties:

- (Compatibility of inverse with source and target)

$$\forall g \in \mathcal{G}_1 \qquad s(g^{-1}) = t(g) \quad \text{ and } t(g^{-1}) = s(g)$$

- (Inverse law)

$$\forall g \in \mathcal{G}_1 \qquad g^{-1} \circ g = u(s(g)) \quad \text{ and } \quad g \circ g^{-1} = u(t(g))$$

2.3. **Examples of groups and groupoids.** In this section we will lay out some of the most basic examples of groups and groupoids. Of particular importance to us will be the action groupoid. Action groupoids can be used to construct some of the most basic non-trivial examples of double structures.

Example 2.5 (Groups)**.** Suppose G is a group. Let $\{*\}$ be a set with a single point and take $s\colon G \to \{*\}$ and $t\colon G \to \{*\}$ to be the unique functions. Take $u\colon \{*\} \to G$ to be the constant map which sends $*$ to the neutral element. If we take i and m to be the usual inverse and multiplication maps for the group then we get the structure of a groupoid $G \rightrightarrows \{*\}$.

Example 2.6 (Group Actions)**.** Suppose G is a group acting on a set X. Let $e \in G$ denote the neutral element and denote the action by $(g, x) \mapsto g \cdot x$.

We can construct a groupoid structure

$$G \times X \rightrightarrows X$$

The source and target maps are as follows:

$$s(g, x) = x \qquad t(g, x) = g \cdot x$$

The unit and inverse maps are:

$$u(x) = (e, x) \qquad i(g, x) = (g^{-1}, g\cdot)$$

Finally, the multiplication map is:

$$(g_1, g_2 \cdot x) \circ (g_2, x) = (g_1 \cdot g_2, x)$$

Example 2.7 (Equivalence relations)**.** Suppose $E \subset X \times X$ is an equivalence relation on a set X. Then we can form a groupoid $E \rightrightarrows X$. The source and target maps are:

$$s(x, y) = y \qquad t(x, y) = x$$

the unit and inverse maps are:

$$u(x) = (x, x) \qquad i(x, y) = (y, x)$$

and the multiplication map is:

$$(x, y) \circ (y, z) = (x, z)$$

In the case that $E = X \times X$ then this is called the *pair groupoid*.

2.4. **Lie groupoids.** Lie groupoids are just groupoids equipped with smooth structure. The literature on Lie groupoids is fairly rich and we will only cover a few of the most basic concepts. For a more thorough reference we refer to Crainic and Fernandes [3] or Mackenzie[6].

Definition 2.8 (Lie groupoid)**.** A *Lie groupoid* is a groupoid $\mathcal{G} \rightrightarrows M$ where the sets $\mathcal{G}$ and M are equipped with second countable smooth manifold structures. We further require that the source and target maps are submersions and the unit, multiplication, and inverse maps are smooth and that M is Hausdorff.

Lie groups have some particular features that are worth point out. One of them is that for each point $x \in M$ in the object manifold, the associated source fiber $s^{-1}(x) \subset \mathcal{G}$ is an embedded submanifold. Even though we do not assume that $\mathcal{G}$ is Hausdorff, such source fibers of $\mathcal{G}$ are automatically Hausdorff.

Many of the examples of groupoids we provided earlier can be made into examples of Lie groupoids. For example, if a Lie group act smoothly on a manifold, then the associated action groupoid is a Lie groupoid. Groupoids associated to equivalence relations are Lie groupoids as long as the equivalence relation $E \subset M \times M$ is an immersed submanifold and it is transverse to the fibers of each of the projection maps $\mathrm{pr}_1, \mathrm{pr}_2 \colon M \times M \to M$. Any countable groupoid $\mathcal{G}$ can be regarded as Lie groupoid under the discrete topology.

3. Double structures

For our purposes, a double structure occurs when a set is equipped with more than one way to multiply elements. There are a few different contexts where this can occur and we will look at a few interesting examples.

3.1. **Compatible operations.** Let us consider the most basic type of double structure:

Definition 3.1 (Compatible binary operations)**.** Suppose S is a set. Let $M_\bullet$ and $M_\circ$ be subsets of $M \times M$ and suppose we have two binary operations:

$$S \times S \supset M_\bullet \to S \qquad (a, b) \mapsto a \bullet b$$

$$S \times S \supset M_\circ \to S \qquad (a, b) \mapsto a \circ b$$

We say that these binary operations are *compatible* if they satisfy the following equation whenever both sides of the equation are well-defined:

$$(a \circ b) \bullet (c \circ d) = (a \bullet c) \circ (b \bullet d) \qquad \text{(Compatibility Law)} \tag{3.1}$$

As we mentioned in the introduction, the Eckmann-Hilton lemma tells us that if a pair of compatible binary operations are unital and globally defined, then they are commutative, associative, and equal.

Example 3.2 (Trivial compatible pair)**.** Suppose $\circ$ is an associative and commutative binary operation. Then $\circ$ with itself constitutes a compatible pair.

Example 3.3. [Matrix multiplication and tensor product] Let

$$\mathrm{Mat} = \bigsqcup_{n,m\in\mathbb{N}} M_{n\times m}(\mathbb{R})$$

be the set of all real matrices of arbitrary dimensions.

Let $\bullet$ be the binary operation arising from matrix multiplication. Note that this binary operation is not globally defined since it requires that the dimensions of the matrices line up appropriately. In fact, this is the composition operation in a category where Mat is the arrows and the objects are natural numbers. The source and target maps are just the dimension maps.

Now, let $\otimes$ be the tensor product (i.e. the Kronecker product of matrices):

$$A \in M_{n\times m},\ B \in M_{p\times q} \qquad (A\otimes B)_{(i\otimes x)(j\otimes y)} := A_{ij}B_{xy}$$

Where we define:

$$i\otimes x := (i-1)p + x \qquad j\otimes y := (j-1)q + y$$

The operation $\otimes$ is globally defined for all matrices and it constitutes a compatible pair with $\bullet$. Since $\bullet$ is not globally defined the Eckmann-Hilton lemma does not apply.

3.2. **Double categories.** Before we can state the definition of a double category, let us make clear the definition of a *morphism of categories*, otherwise known as a functor.

Definition 3.4. Suppose $\mathcal{C}$ and $\mathcal{D}$ are categories. We will denote the structure maps the same way and leave the distinction implicit from the domains. A functor $\phi\colon \mathcal{C}\to\mathcal{D}$ consists of a pair of functions

$$\phi_1\colon \mathcal{C}_1\to\mathcal{D}_1 \qquad \phi_0\colon \mathcal{C}_0\to\mathcal{D}_0$$

with the following properties:

- (Compatibility with source and target)

$$s\circ\phi_1 = \phi_0\circ s \qquad t\circ\phi_1 = \phi_0\circ t$$

- (Compatibility with multiplication)

$$\forall(a,b)\in\mathcal{C}_1\times_{s,t}\mathcal{C}_1\to\mathcal{C}_1 \qquad \phi_1(a\bullet b) = \phi_1(a)\bullet\phi_1(b)$$

A functor can be visualized as a square:

$$\begin{array}{ccc} \mathcal{C}_1 & \xrightarrow{\phi_1} & \mathcal{D}_1 \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{C}_0 & \xrightarrow{\phi_0} & \mathcal{D}_0 \end{array}$$

A double category is a category internal to the category of categories. This is not the most useful definition for those who do not already know what such structures are.

Definition 3.5. A double category $\mathcal{D}$ consists of four sets $\mathcal{C}$, $\mathcal{K}$, $\mathcal{H}$ and M together with four category structures which we visually arrange into a square:

$$\begin{array}{ccc} \mathcal{C} & \rightrightarrows & \mathcal{K} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{H} & \rightrightarrows & M \end{array}$$

The various structures are assumed to be *compatible* in the sense that the source and target maps of $\mathcal{C} \rightrightarrows \mathcal{H}$ and $\mathcal{H} \rightrightarrows M$ constitute functors from $\mathcal{C} \rightrightarrows \mathcal{K}$ to $\mathcal{H} \rightrightarrows M$:

$$\begin{array}{ccc} \mathcal{C} & \xrightarrow{s_1} & \mathcal{K} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{H} & \xrightarrow{s_0} & M \end{array} \qquad \begin{array}{ccc} \mathcal{C} & \xrightarrow{t_1} & \mathcal{K} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{H} & \xrightarrow{t_0} & M \end{array}$$

The two multiplication operations on $\mathcal{G}$ must also constitute a pair of compatible binary operations. This is the algebraic rule encoding that the composition operation also constitutes a homomorphism. If $\circ$ is the binary operation for $\mathcal{C} \rightrightarrows \mathcal{K}$ and $\bullet$ is the binary operation for $\mathcal{C} \rightrightarrows \mathcal{H}$ then for any four elements $a, b, c, d \in \mathcal{C}$ such that the following operations are well-defined:

$$a \circ b, \quad c \circ d, \quad a \bullet c, \quad b \bullet d \tag{3.2}$$

we must have that the following equation is also well-defined and holds:

$$(a \circ b) \bullet (c \circ d) = (a \bullet c) \circ (b \bullet d) \tag{3.3}$$

Generally speaking, there are quite a large number of structure maps associated to a double category. We will generally try to avoid using them explicitly. Most of the time we will be concerned with the compatible pair of binary operations and we will often use the phrase "whenever it is well defined" as a shorthand for the appropriate composition rules.

However, in situations where we must make reference to these structures, we will observe the following convention: We call the category structures on $\mathcal{C} \rightrightarrows \mathcal{H}$ and $\mathcal{K} \rightrightarrows M$ the *horizontal categories* and the category structure maps will be denoted:

$$s_i^H,\ t_i^H,\ u_i^H,\ m_i^H,\ \bullet_i \qquad i = 0, 1$$

where $i = 1$ corresponds to $\mathcal{C} \rightrightarrows \mathcal{H}$ and $i = 0$ corresponds to $\mathcal{K} \rightrightarrows M$.

The category structures on $\mathcal{C} \rightrightarrows \mathcal{K}$ and $\mathcal{H} \rightrightarrows M$ are the *vertical categories* and the category structure maps will be denoted:

$$s_i^V,\ t_i^V,\ u_i^V,\ m_i^V,\ \circ_i \qquad i = 0, 1$$

where $i = 1$ corresponds to $\mathcal{C} \rightrightarrows \mathcal{K}$ and $i = 0$ corresponds to $\mathcal{H} \rightrightarrows M$.

It is often helpful to visualize elements $a \in \mathcal{C}$ as a square:

$$\begin{array}{ccc} x & \xrightarrow{s_1^V(a)} & y \\ {\scriptstyle s_1^H(a)}\downarrow & a & \downarrow{\scriptstyle t_1^H(a)} \\ z & \xrightarrow[t_1^V(a)]{} & w \end{array}$$

where x, y, z, w are the sources and targets of their respective arrows. The elements of G can be thought of as 2-cells and we think of the subscripts 1 as correspond to passing to "1-cells" while the subscripts 0 correspond to passing to "0-cells."

The vertical and horizontal composition operations can be visualized by vertically or horizontally juxtaposing such squares. In this way, the compatibility law looks like:

$$\begin{pmatrix} a \\ \circ \\ b \end{pmatrix} \bullet \begin{pmatrix} c \\ \circ \\ d \end{pmatrix} = \begin{matrix} (a & \bullet & c) \\ & \circ & \\ (b & \bullet & d) \end{matrix}$$

3.3. **Double groupoids.** Double groupoids are a type of double category that is of particular interest to us.

Definition 3.6. A double groupoid is a double category:

$$\begin{array}{ccc} \mathcal{G} & \rightrightarrows & \mathcal{K} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{H} & \rightrightarrows & M \end{array}$$

where all of the category structures are in fact groupoids.

Let us consider a few examples.

Example 3.7 (Groups)**.** Let G be a group. Consider the following double groupoid:

$$\begin{array}{ccc} G & \rightrightarrows & G \\ \downarrow\downarrow & & \downarrow\downarrow \\ \{*\} & \rightrightarrows & \{*\} \end{array}$$

For the top and bottom groupoid structures, we take the trivial groupoid structure in which every arrow is a unit. For the left and right groupoid structures, we take the usual group operations. It is straightforward to check that these structures are compatible.

Example 3.8 (Strict 2-groupoids)**.** A strict 2-groupoid is a kind of double groupoid where the bottom groupoid structure is trivial:

$$\begin{array}{ccc} \mathcal{G}_2 & \rightrightarrows & \mathcal{G}_1 \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{G}_0 & \rightrightarrows & \mathcal{G}_0 \end{array}$$

Under the typical conventions for 2-categories, the set $\mathcal{G}_0$ is the *objects*, $\mathcal{G}_1$ is the *arrows* and $\mathcal{G}_2$ is the *2-arrows*.

Example 3.9 (Strict 2-groups)**.** A strict 2-group is a 2-groupoid with a single object. Hence, it can be interpreted as a double groupoid of the form:

$$\begin{array}{ccc} G_2 & \rightrightarrows & G_1 \\ \downarrow\downarrow & & \downarrow\downarrow \\ \{*\} & \rightrightarrows & \{*\} \end{array}$$

Lie double groupoids arise when we impose smoothness conditions on the structure maps. However, there is one slight caveat.

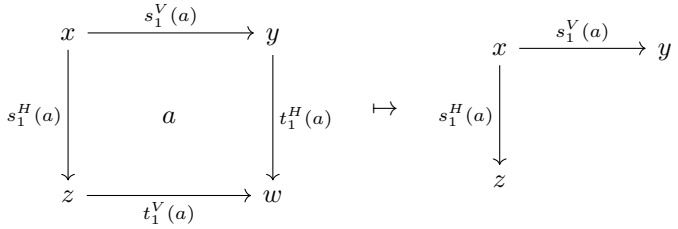
Definition 3.10. A *Lie double groupoid* is a double groupoid:

$$\begin{array}{ccc} \mathcal{G} & \rightrightarrows & \mathcal{K} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{H} & \rightrightarrows & M \end{array}$$

where $\mathcal{G}, \mathcal{K}, \mathcal{H}, M$ are all equipped with smooth manifold structures which make the associated groupoid structures into Lie groupoids. We additionally require that the following map is a surjective submersion:

$$s_1^V \times s_1^H : \mathcal{G} \to \{(k,h) \in \mathcal{K} \times \mathcal{H} \ : \ s_0^V(k) = s_0^H(h)\} \qquad a \mapsto (s_1^V(a), s_1^H(a))$$

The map $s_1^V \times s_1^H$ is called the *double source map* of the groupoid. In terms of squares, it can be visualized as the operation:



From this picture it is clear that it is a kind of smooth filling condition. Practically speaking, the significance of requiring that the double source map be a surjective submersion is that it ensures that the sets of vertically composable arrows in $\mathcal{G}$ is a *smooth* Lie subgroupoid of the horizontal groupoid structure on $\mathcal{G} \times \mathcal{G}$ (and vice versa). It also ensures that the space of composable squares $\mathcal{G}^{\square}$, which is the domain of the compatibility law, is a smooth manifold:

$$\mathcal{G}^{\square} := \{(a,b,c,d) \in \mathcal{G} \ : \ a \circ b, \ c \circ d, \ a \bullet c, \ b \bullet d \text{ well-defined}\}$$

4. Algebras associated to categories

4.1. **Category algebras.** The simplest sort of algebra that we can attach to a category is the so-called category algebra.

Definition 4.1. [Category algebra] Let $\mathcal{C}$ be a category. The *category algebra* of $\mathcal{C}$ is the vector space $\mathbb{C}\mathcal{C}$ generated freely by elements of $\mathcal{C}$.

Associated to any element $a \in \mathcal{C}_1$ let $\delta_a \in \mathbb{C}\mathcal{C}$ denote the associated basis vector. The algebra structure on $\mathbb{C}\mathcal{C}$ is defined in terms of basis elements:

$$\forall a, b \in \mathcal{C}_1 \qquad \delta_a * \delta_b := \begin{cases} \delta_{a \bullet b} & s(a) = t(b) \\ 0 & s(a) \neq t(b) \end{cases}$$

If $\mathcal{C}$ is a groupoid this is called the *groupoid algebra*. In the case where $\mathcal{C}$ is a group, then this is the classical group algebra.

One of the main distinctions between a category and its associated category algebra is that the product operation is globally defined.

Example 4.2. Suppose S is a finite set with n elements and $\mathcal{G} = S \times S \rightrightarrows S$ is the pair groupoid. Then the group algebra of $\mathcal{G}$ is isomorphic to $M_{n\times n}(\mathbb{C})$ the algebra of n by n matrices.

To see why, let us chose a way to index S by natural numbers: $S = \{s_i\}_{1\leq i\leq n}$. Then consider the isomorphism given by the following map of basis elements:

$$\forall\ 0 \leq i, j \leq n \qquad \delta_{(s_i,s_j)} \mapsto E_{ij}$$

where E_{ij} is the elementary $n \times n$ matrix with a single 1 in the (i, j)th entry. We leave it to the reader to verify that this is indeed an isomorphism of algebras.

4.2. **Convolution algebras.** The convolution algebra of a topological groupoid provides us with a simultaneous generalization of the notion of a group algebra and the algebra of continuous functions on a space. A more thorough reference on the basics of Haar systems and convolution algebras in groupoids can be found in Williams [10]. We will stick to the setting of Lie groupoids but much of the following discussion is well -defined for many (nice enough) topological groupoids by replacing "smooth" with "continuous." Throughout this section, given a manifold M, we will use the notation $C_c^\infty(M)$ to denote the set of complex valued, compactly supported, smooth functions on M.

Before we can define the convolution algebra we require the groupoid analogue of a Haar measure.

Definition 4.3 (Haar system)**.** Suppose $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ is a Lie groupoid. A *smooth (left) Haar system* on $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ is an assignment to each $x \in \mathcal{G}_0$ a Radon measure μ_x on $t^{-1}(x)$ with the following properties:

(1) **Smoothness**: For each function $u \in C_c^\infty(\mathcal{G}_1)$ the associated function:

$$\mathcal{G}_0 \to \mathbb{R} \qquad x \mapsto \int\limits_{t^{-1}(x)} (u|_{t^{-1}(x)}) d\mu_x$$

is smooth.

(2) **Left Invariant**: For each $g \in \mathcal{G}_1$ with $s(g) = x$ and $t(g) = y$ the function:

$$L_g \colon t^{-1}(x) \to t^{-1}(y) \qquad h \mapsto g \circ h$$

is measure preserving.

Example 4.4. Suppose G is a Lie group. If we consider G to be a groupoid with object space $\{*\}$ then a Haar system on $G \rightrightarrows \{*\}$ is the same thing as a Haar measure on G.

Example 4.5. Suppose $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ is an étale Lie groupoid. In other words, the source and target maps are étale maps. Then the t-fibers of $\mathcal{G}_1$ are zero dimensional and any scalar multiple of the counting measure on the t-fibers constitutes a Haar system.

Definition 4.6. Suppose $u, v \in C_c^\infty(\mathcal{G}_1)$. Then the *convolution* of u with v is the function:

$$u * v \colon \mathcal{G}_1 \to \mathbb{C} \qquad u * v(g) := \int\limits_{t^{-1}(t(g))} u(h) v(h^{-1} \circ g)\, d\mu_{t(g)}(h).$$

This defines a (possibly non-unital) associative algebra structure:

$$C_c^\infty(\mathcal{G}_1) \otimes C_c^\infty(\mathcal{G}_1) \to C_c^\infty(\mathcal{G}_1).$$

There is an alternative construction of the convolution operation that provides us with a bit more insight than the usual formula. To start, we first observe that given $g \in \mathcal{G}_1$ we can canonically identify $m^{-1}(g)$ with $t^{-1}(t(g))$ by taking the map:

$$t^{-1}(t(g)) \to m^{-1}(g) \qquad h \mapsto (h, h^{-1} \circ g)$$

This provides us with a smooth family of measures μ_g on $m^{-1}(g)$ for each $g \in \mathcal{G}_1$. From this point of view, the convolution of u with v is obtained by viewing $u \otimes v$ as a function on $\mathcal{G}_1 \times \mathcal{G}_1$, restricting it to the set of composable arrows, and integrating along the fibers of m:

$$C_c^\infty(\mathcal{G}_1) \otimes C_c^\infty(\mathcal{G}_1) \longrightarrow C_c^\infty(\mathcal{G}_1 \times \mathcal{G}_1) \xrightarrow{\text{restrict}} C_c^\infty(\mathcal{G}_1 \times_{s,t} \mathcal{G}_1) \xrightarrow{\int d\mu_g} C_c^\infty(\mathcal{G}_1)$$

Example 4.7. Suppose $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ has the discrete topology. Take the Haar system on $\mathcal{G}_1 \rightrightarrows \mathcal{G}_0$ which is given by the counting measure. The set $C_c^\infty(\mathcal{G}_1)$ can be canonically identified with $\mathbb{C}\mathcal{G}_1$ and this constitutes an isomorphism of the convolution algebra with the groupoid algebra.

Example 4.8. Consider $\mathbb{R} \rightrightarrows \{*\}$ where we take the group structure on $\mathbb{R}$ by addition. We can equip $\mathbb{R}$ with the standard measure. Then the convolution of a function u with a function v is the classical convolution:

$$u * v(x) = \int_{\mathbb{R}} u(y)v(x-y)dy.$$

Example 4.9. Suppose K is a discrete group acting smoothly on a manifold M. The action groupoid $K \ltimes M \rightrightarrows M$ is an étale groupoid. Indeed, the source and target fibers of $K \ltimes M$ can be identified with K. Let us take the counting measure as the Haar system on $K \ltimes M \rightrightarrows M$. Since our measure is discrete, integration over the target fibers is given by a sum.

In this case, convolution is given by the formula:

$$u * v(k, p) := \sum_{h \in K} u(h, (h^{-1}k) \cdot p)v(h^{-1}k, p).$$

This sum is finite so long as u and v are compactly supported.

5. Convolution algebras of double groupoids

With the background out of the way, we will now introduce the basic problem. Our starting point will be a double groupoid:

$$\begin{array}{ccc} \mathcal{G} & \rightrightarrows & \mathcal{K} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathcal{H} & \rightrightarrows & M \end{array}$$

Let us denote by $\circ$ the groupoid operation for $\mathcal{G} \rightrightarrows \mathcal{K}$ and $\bullet$ the groupoid operation for $\mathcal{G} \rightrightarrows \mathcal{H}$.

If we choose Haar systems for each of these groupoid structures, we obtain two algebra structures on $C_c^\infty(\mathcal{G})$ which we denote by $\overset{\circ}{*}$ and $\overset{\bullet}{*}$, respectively. Recall that, for a suitably composable quadruple of elements, $a, b, c, d \in \mathcal{G}$ we have that an intertwining law holds:

$$(a \circ b) \bullet (c \circ d) = (a \bullet c) \circ (b \bullet d).$$

Our goal will be to compute a few examples to see how the intertwining law plays into the structure of the convolution algebra.

We noted in the introduction that the Eckmann-Hilton lemma suggests that this formula should not hold in general. By looking at some examples, it is not too difficult to see that this is the case.

5.1. **Countable case.** Let us suppose that $\mathcal{G}$ is countable so it is equipped with the discrete topology. We saw in the previous section (Example 4.7) that $C_c^\infty(\mathcal{G})$ can be identified with the freely generated vector space $\mathbb{C}\mathcal{G}$. Furthermore, under this isomorphism the convolution operations $\overset{\bullet}{*}$ and $\overset{\circ}{*}$ are just the corresponding groupoid algebra products.

Our first observation is just a slight modification of the Eckmann-Hilton lemma.

Proposition 5.1. *Suppose $\mathcal{G}$ is a countable double groupoid. Then the convolution operations $\overset{\bullet}{*}$ and $\overset{\circ}{*}$ constitute a compatible pair of binary operations if and only if the two groupoid products are equal, $\bullet = \circ$.*

Proof. Since $\mathcal{G}$ is countable, let us index it by natural numbers for convenience:

$$\mathcal{G} = \{g_i\}_{i\in\mathbb{N}}$$

Now consider the subset $U^V \subset \mathcal{G}$ of elements which are units with respect to the vertical structure. Similarly let $U^H \subset \mathcal{G}$ be the subset of units with respect to the horizontal structure.

Now, for natural numbers $n \in \mathbb{N}$, consider the following functions on $\mathcal{G}$:

$$e_n^V : \mathcal{G} \to \mathbb{C} \qquad e_n^V(g) := \sum_{g_i \in U^V}^{i<n} \delta_{g_i}$$

$$e_n^H : \mathcal{G} \to \mathbb{C} \qquad e_n^H(g) := \sum_{g_i \in U^H}^{i<n} \delta_{g_i}$$

In other words, e_n^V is the sum of the delta functions for first n elements of $\mathcal{G}$ that are vertical units. Similarly for e_m^H.

Note that the product of two delta functions is always either zero, or a delta function for the same element. Therefore it follows that

$$e_n^V \overset{\bullet}{*} e_m^H = \sum_{g_i \in U^V \cap U^H}^{i<min(m,n)} \delta_{g_i}$$

In other words, the convolution of e_n^V with e_M^H is just a sum of delta functions which are units with respect to both structures.

Furthermore, observe that for all $u \in C_c^\infty(\mathcal{G})$, there exists a natural number n such that:

$$e_n^V \overset{\circ}{*} u = u \overset{\circ}{*} e_n^V = u$$

Similarly for e_m^H and $\overset{\bullet}{*}$.

Now consider the equation:

$$(e_n^V \overset{\circ}{*} e_m^H) \overset{\bullet}{*} (e_m^H \overset{\circ}{*} e_n^V) = (e_n^V \overset{\bullet}{*} e_m^H) \overset{\circ}{*} (e_m^H \overset{\bullet}{*} e_n^V). \tag{5.1}$$

From our previous observation, the result of the above computation will be a sum of delta functions for elements which are units with respect to both product structures.

On the other hand, if n is large enough relative to m we have that the left hand side of Equation 5.1 becomes:

$$(e_n^V \overset{\circ}{*} e_m^H) \overset{\bullet}{*} (e_m^H \overset{\circ}{*} e_n^V) = e_m^H \overset{\bullet}{*} e_m^H = e_m^H$$

This implies that every one of the constituent delta functions of e_m^H comes from a unit for the vertical composition. A symmetrical argument coming from computing the right hand side of Equation 5.1 implies that each of the delta functions for e_n^V come from elements that are units with respect to horizontal composition. In other words, the units for vertical and horizontal convolutions are the same and so:

$$e_n := e_n^H = e_n^V$$

To finish the proof, we now consider that for a natural number n and element $u, v \in C_c^\infty(\mathcal{G})$ we have that:

$$(u \overset{\circ}{*} e_n) \overset{\bullet}{*} (e_n \overset{\circ}{*} v) = (u \overset{\bullet}{*} e_n) \overset{\circ}{*} (e_n \overset{\bullet}{*} v).$$

For n sufficiently large it follows that:

$$u \overset{\bullet}{*} v = (u \overset{\circ}{*} e_n) \overset{\bullet}{*} (e_n \overset{\circ}{*} v) = (u \overset{\bullet}{*} e_n) \overset{\circ}{*} (e_n \overset{\bullet}{*} v) = u \overset{\circ}{*} v.$$

Hence $\overset{\bullet}{*}$ and $\overset{\circ}{*}$ are equal. Since $\overset{\circ}{*}$ and $\overset{\bullet}{*}$ are groupoid algebras, by computing $\overset{\circ}{*}$ and $\overset{\bullet}{*}$ on basis elements it follows that $\bullet = \circ$. □

We consider it likely that a version of the above proof exists for convolution algebras of double Lie groupoids. However, we will not include a proof of the more general case here.

6. Noncommutative torus

We will now take a look at a slightly more complicated but quite important example: the noncommutative torus. Let $r \in \mathbb{R}$ be a fixed real number. We consider the circle group $\mathbb{S}^1$ as a quotient of $\mathbb{R}$ by the subgroup $2\pi\mathbb{Z}$. Now, consider the group homomorphism:

$$\phi_r \colon \mathbb{Z} \to \mathbb{S}^1 \qquad \phi_r(n) = [rn]_{2\pi}$$

With this data, one can construct a double groupoid (indeed a 2-group) of the form:

$$\begin{array}{ccc} \mathbb{Z} \ltimes \mathbb{S}^1 & \rightrightarrows & \mathbb{S}^1 \\ \downarrow\downarrow & & \downarrow\downarrow \\ \{*\} & \rightrightarrows & \{*\} \end{array}$$

As a set, $\mathbb{Z} \ltimes \mathbb{S}^1$ is just the standard Cartesian product. The semi-direct product notation is used due to the fact that the groupoid structure on $\mathbb{Z} \ltimes \mathbb{S}^1 \rightrightarrows \mathbb{S}^1$ is the action groupoid associated to the homomorphism $\phi_r \colon \mathbb{Z} \to \mathbb{S}^1$.

In other words, the source and target are given by

$$s(n, \theta) = \theta \qquad t(n, \theta) = rn + \theta$$

If we have (m, ψ) and (n, θ) composable, then:

$$(m, \psi) \circ (n, \theta) = (n + m, \psi)$$

Finally, the inverse map is:

$$i(n, \theta) = (-n, rn + \theta).$$

The group structure on $\mathbb{S}^1 \rightrightarrows \{*\}$ is just the standard circle group and the group structure on $\mathbb{Z} \ltimes \mathbb{S}^1 \rightrightarrows \{*\}$ is the one obtained by regarding it as a simple Cartesian product of groups.

As usual we will refer to $\circ$ as the *vertical composition* and $\bullet$ as the horizontal composition.

We can roughly summarize the calculation we do in this section as follows: If we apply a fourier transform to $\mathbb{Z} \ltimes \mathbb{S}^1$ we can obtain an isomorphism (up to closure) between the convolution algebras of a double groupoid of the form:

$$\begin{array}{ccc} \mathbb{Z} \times \mathbb{Z} & \rightrightarrows & \{*\} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathbb{Z} & \rightrightarrows & \{*\} \end{array}$$

and the convolution algebras of:

$$\begin{array}{ccc} \mathbb{Z} \ltimes \mathbb{S}^1 & \rightrightarrows & \mathbb{S}^1 \\ \downarrow\downarrow & & \downarrow\downarrow \\ \{*\} & \rightrightarrows & \{*\} \end{array}$$

The groupoid structure $\mathbb{Z} \times \mathbb{Z} \rightrightarrows \mathbb{Z}$ is the one obtained by regarding it as a trivial bundle of groups over $\mathbb{Z}$. Under this correspondence, the compatibility condition for convolution holds on generators which are "composable" relative to this groupoid structure. See Subsection 8.1 for further details about the Fourier groupoid.

6.1. **Orthonormal basis.** The space of functions on $\mathbb{S}^1$, with Haar measure μ that has been normalized (so that $\mu(\mathbb{S}^1) = 1$), has a particularly nice form. In particular, it admits a nice countable basis indexed by the integers. Since $\mathbb{Z} \ltimes \mathbb{S}^1$ is just the product space, any compactly supported function $f \in C_c^\infty(\mathbb{Z} \ltimes \mathbb{S}^1)$ can be written as a sum of functions $g \cdot h$ where $g \in C_c^\infty(\mathbb{Z})$ and $h \in C^\infty(\mathbb{S}^1)$. We will take advantage of this basis to investigate the relationship between $\overset{\bullet}{*}$ and $\overset{\circ}{*}$.

For each $k \in \mathbb{Z}$, let $e_k(\theta) = e^{ik\theta}$. The collection of functions $\{e_k\}_{k\in\mathbb{Z}}$ constitutes and orthonormal (relative to the L^2 inner product) basis for $C^\infty(\mathbb{S}^1)$.

We can use these functions to construct a related basis for $C_c^\infty(\mathbb{Z} \ltimes \mathbb{S}^1)$. Consider the collection of functions $\{u_{jk}\}_{j,k\in\mathbb{Z}} \subset C_c^\infty(\mathbb{Z} \ltimes \mathbb{S}^1)$ where:

$$u_{jk}(n, \theta) := \begin{cases} e^{ik\theta} & \text{if } n = j \\ 0 & \text{otherwise.} \end{cases} \tag{6.1}$$

In other words the index j refers to the level of $\mathbb{Z}$ where u_{jk} is supported and the index k refers to the frequency of u_{jk}.

6.2. **Convolution algebras.** Note that the source fibers of $\mathbb{Z} \ltimes \mathbb{S}^1 \rightrightarrows \mathbb{S}^1$ are discrete, so for our vertical Haar system μ^V, we simply take the counting measure. For a Haar measure on $\mathbb{Z} \ltimes \mathbb{S}^1 \rightrightarrows \{*\}$ we simply take μ^H to be the product of the counting measure on $\mathbb{Z}$ with the normalized Haar measure on $\mathbb{S}^1$.

Utilizing the definition of the convolution algebras yields that, for any pair of compactly supported functions $u, v \in C_c^\infty(\mathbb{Z} \ltimes \mathbb{S}^1)$, we have:

$$u \overset{\bullet}{*} v(n,\theta) = \sum_{m\in\mathbb{Z}} \int_{\phi\in\mathbb{S}^1} u(m,\phi)v(n-m,\theta-\phi)\, d\phi, \tag{6.2}$$

$$u \overset{\circ}{*} v(n,\theta) = \sum_{m\in\mathbb{Z}} u(m, r(n-m)+\theta)v(n-m,\theta). \tag{6.3}$$

Recall that the real number $r \in \mathbb{R}$ refers to the constant determining the action of $\mathbb{Z}$ on $\mathbb{S}^1$. Note that one of these convolutions includes an integral while the other only includes a sum. Since we assume that u and v are compactly supported, the sums are finite.

Proposition 6.1. *Let u_{ab} and u_{cd} be orthonormal basis as defined in* (6.1)*. Then*

$$u_{ab} \overset{\circ}{*} u_{cd} = e^{irbc} u_{(a+c)(b+d)}. \tag{6.4}$$

Proof. We have

$$\begin{aligned} u_{ab} \overset{\circ}{*} u_{cd}(n,\theta) &= \sum_{m\in\mathbb{Z}} u_{ab}(m, r(n-m)+\theta)u_{cd}(n-m),\theta) \\ &= u_{ab}(a, r(n-a)+\theta)u_{cd}(n-a),\theta). \end{aligned} \tag{6.5}$$

Note that if $c \neq n-a$ then the expression (6.5) just becomes zero. If $c = n-a$, then the expression (6.5) becomes

$$e^{irbc} e^{ib\theta} e^{id\theta} = e^{irbc} e^{i(b+d)\theta}.$$

So

$$u_{ab} \overset{\circ}{*} u_{cd}(n,\theta) = \begin{cases} e^{irbc} u_{(a+c)(b+d)} & \text{if } n = a+c \\ 0 & \text{otherwise,} \end{cases}$$

thus proving (6.4) □

The above calculation is the standard method for recovering the classical "noncommutative torus" algebra out of the action groupoid associated to a homomorphism $\mathbb{Z} \to \mathbb{S}^1$.

Now we compute the other convolution in this basis.

Proposition 6.2. *Let u_{ab} and u_{cd} be orthonormal basis as defined in* (6.1)*. Then*

$$u_{ab} \overset{\bullet}{*} u_{cd} = \begin{cases} u_{(a+c)b} & \textit{if } b = d \\ 0 & \textit{otherwise.} \end{cases} \tag{6.6}$$

Proof. We have

$$\begin{aligned} u_{ab} \overset{\bullet}{*} u_{cd}(n,\theta) &= \sum_{m\in\mathbb{Z}} \int_{\phi\in\mathbb{S}^1} u_{ab}(m,\phi)u_{cd}(n-m,\theta-\phi)\, d\phi. \\ &= \int_{\phi\in\mathbb{S}^1} u_{ab}(a,\phi)u_{cd}(n-a,\theta-\phi)\, d\phi. \end{aligned} \tag{6.7}$$

If $c = n-a$ (thus $n = a+c$) then the integral (6.7) becomes

$$\int_{\mathbb{S}^1} e^{ib\phi} e^{id(\theta-\phi)}\, d\phi. \tag{6.8}$$

If $b = d$ then (6.8) becomes

$$\begin{aligned} & e^{id\theta} \int_{\mathbb{S}^1} 1 \, d\phi \\ =& e^{id\theta} \\ =& u_{(a+c)d}(n, \theta). \end{aligned}$$

(since the measure on $\mathbb{S}^1$ has been normalized). Otherwise (if $b \neq d$ or $n \neq a + c$), we have zero. □

6.3. **Compatibility behavior of the two convolutions.** We observed earlier that for countable double groupoids, the compatibility of the convolution products appears to depend on multiplying "composable" elements. We will see that, under our basis, composability is determined by the *frequency* component.

We consider the following two expressions

$$(u_{ab} \overset{\circ}{*} u_{cd}) \overset{\bullet}{*} (u_{ef} \overset{\circ}{*} u_{gh}) \tag{6.9}$$

and

$$(u_{ab} \overset{\bullet}{*} u_{ef}) \overset{\circ}{*} (u_{cd} \overset{\bullet}{*} u_{gh}). \tag{6.10}$$

Proposition 6.3. *The two expressions* (6.9) *and* (6.10) *are equal whenever* $b = f$ *and* $d = h$*. It also holds whenever* $b + d \neq f + h$*.*

Proof. The expression (6.9) gives us

$$\begin{aligned} & \left(e^{irbc} u_{(a+c)(b+d)}\right) \overset{\bullet}{*} \left(e^{irfg} u_{(e+g)(f+h)}\right) \\ =& e^{r(bc+fg)} \left(u_{(a+c)(b+d)} \overset{\bullet}{*} u_{(e+g)(f+h)}\right) \\ =& \begin{cases} e^{ir(bc+fg)} u_{(a+c+e+g)(b+d)} & \text{if } b + d = f + h \\ 0 & \text{otherwise.} \end{cases} \end{aligned}$$

On the other hand, if $b \neq f$ or $d \neq h$, then (6.10) is just zero. So if $b = f$ and $d = h$, then the expression (6.10) becomes

$$\begin{aligned} & u_{(a+e)b} \overset{\circ}{*} u_{(c+g)d} \\ =& e^{ib(c+g)} u_{(a+e+c+g)(b+d)} \end{aligned}$$

That is, expression (6.10) is

$$\begin{cases} e^{ib(c+g)} u_{(a+e+c+g)(b+d)} & \text{if } b = f \text{ and } d = h \\ 0 & \text{otherwise.} \end{cases}$$

From here, it is easy to see that expression (6.9) becomes (6.10) if $b = f$ and $d = h$. We also see that (6.9) and (6.10) are zero whenever $b + d \neq f + h$. □

6.4. **As a countable double groupoid.** It is somewhat surprising that the composability rule that permits the compatibility law to hold is based on the frequency component. One way to explain this is that the basis we are using is obtained via a fourier transform.

Consider the following double groupoid:

$$\begin{array}{ccc} \mathbb{Z}\times\mathbb{Z} & \rightrightarrows & \{*\} \\ \downarrow\downarrow & & \downarrow\downarrow \\ \mathbb{Z} & \rightrightarrows & \{*\} \end{array}$$

where the product on $\mathbb{Z}\times\mathbb{Z}\rightrightarrows\{*\}$ is just the usual Cartesian product and the groupoid structure on $\mathbb{Z}\times\mathbb{Z}\rightrightarrows\mathbb{Z}$ is as follows:

The source and target maps $\mathbb{Z}\times\mathbb{Z}\to\mathbb{Z}$ are both projection to the second component. The product operation on $\mathbb{Z}\times\mathbb{Z}$ is obtained by regarding it as a bundle of groups. In other words, (a,b) and (c,d) are composable if and only if $b=d$ and in that case:

$$(a,b)\overset{\bullet}{*}(c,d)=(a+c,b)=(a+c,d).$$

It is straightforward to check that this is a well-defined double groupoid (see Subsection 8.1 for details about the Fourier groupoid). If we use counting measures, and the standard basis on $C_c^\infty(\mathbb{Z}\times\mathbb{Z})$ in terms of delta functions, there is a natural correspondence between basis elements of $\mathbb{Z}\times\mathbb{Z}$ and our basis $\{u_{ab}\}$ by associating:

$$\delta_{(a,b)} \quad\Leftrightarrow\quad u_{ab}$$

This yields a dense inclusion:

$$C_c^\infty(\mathbb{Z}\times\mathbb{Z})\hookrightarrow C_c^\infty(\mathbb{Z}\ltimes\mathbb{S}^1).$$

Proposition 6.4. *If $r\in 2\pi\mathbb{Z}$ then the inclusion $C_c^\infty(\mathbb{Z}\times\mathbb{Z})\hookrightarrow C_c^\infty(\mathbb{Z}\ltimes\mathbb{S}^1)$ is an isomorphism with respect to both $\overset{\bullet}{*}$ and $\overset{\circ}{*}$.*

Proof. If $r\in 2\pi\mathbb{Z}$ then the constant term e^{rbc} in Equation (6.4) is equal to one. If we rewrite this equation in terms of delta functions we get:

$$\delta_{(a,b)}\overset{\circ}{*}\delta_{(c,d)}=\delta_{((a+c),(b+d))}.$$

which is precisely the group algebra of $\mathbb{Z}\times\mathbb{Z}\rightrightarrows\{*\}$.

Similarly, if we rewrite Equation (6.6) in terms of delta functions:

$$\delta_{(a,b)}\overset{\bullet}{*}\delta_{(c,d)}=\begin{cases}\delta_{((a+c),b)} & \text{if } b=d\\ 0 & \text{otherwise.}\end{cases}$$

which is precisely the groupoid algebra for $\mathbb{Z}\times\mathbb{Z}\rightrightarrows\mathbb{Z}$. □

In the case that r is not an integer multiple of 2π, then the previous lemma is not quite true due to the fact that the products differ for $\overset{\circ}{*}$. In order to extend this lemma to the general case one needs to utilize the notion of a convolution algebra *twisted* by a group action. See Renault [8] for a nice treatment of the topic.

7. Review: Representations of compact Lie groups

All of the remaining calculations we will do will rely on a variety of facts about the representation theory of compact Lie groups. In this section we will review Schur's orthogonality relations, the Peter-Weyl theorem, and some of their basic consequences for computing convolutions. For a more detailed reference, see Sections 1.3 and 1.5 of [5]. Throughout this section G is assumed to be a compact Lie group and the Haar measure on G is taken to be the unique Haar measure μ with the property that $\int_G 1d\mu=1$.

Our final goal is to attempt to generalize our calculations for the case of the noncommutative torus. The correct analogue of the fourier transform in this context will arise out of the study of irreducible representations.

7.1. **Matrix coefficients.** Let us begin by defining the types of representations we are interested in.

Definition 7.1. A *unitary representation* (π, V_π) is a strongly continuous homomorphism $\pi : G \to U(V_\pi)$ from G into unitary operators of a Hilbert space V_π.

Given a unitary representation (π, V_π), we will use d_π to denote the dimension of V_π. We may sometimes write simply π to denote a unitary representation without directly invoking the underlying vector space. However, V_π will always refer to the underlying vector space for a unitary representation π.

Definition 7.2. Given a unitary representation π, if u, v are vectors in V_π, we define the *matrix coefficient for* (u, v) to be

$$\pi_{uv} \colon G \to \mathbb{C} \qquad \pi_{uv}(g) := \langle u, \pi(g)v \rangle$$

We will be using matrix coefficients to construct a basis for functions on G.

Definition 7.3. We say that a representation π is *irreducible* if the only invariant closed subspaces of V_π under the action of G is V itself and the trivial subspace.

Definition 7.4. Two unitary representations π and σ of G are *(unitarily) equivalent* if there is an invertible and unitary *intertwining operator* $J : V_\pi \to V_\sigma$ such that $J\pi(g) = \sigma(g)J$ for all $g \in G$. we define $\widehat{G}$ to be the set of equivalence classes of irreducible unitary representations of G.

The set of irreducible representations, up to unitary equivalence, can be rather complex for a general Lie group. However, in the special case that G is compact, significant simplifications can be made. For instance:

Proposition 7.5. *If G is a compact Lie group, then all irreducible unitary representations of G are finite-dimensional and $\widehat{G}$ is countable.*

The previous fact tells us that, up to unitary equivalence, all unitary representations are equivalent to one where V_π is just Euclidean space with the standard basis and metric. This can be easily seen by choosing an orthormal basis for V_π.

Therefore, from now on let us assume that our irreducible representations (π, V_π) are come equipped with an ordered orthonormal basis. We will use the notation $e_i^{(\pi)}$ where $1 \leq i \leq d_\pi$ to denote the basis vectors. Sometimes, we may omit the superscript π when it is clear from context. Under these conventions, let:

$$1 \leq i, j \leq d_\pi \quad \pi_{ij} \colon G \to \mathbb{C} \qquad \pi_{ij}(g) := \pi_{e_i e_j} = \langle e_i^{(\pi)}, \pi(g) e_j^{(\pi)} \rangle$$

7.2. **Schur's Orthogonality relations and Peter-Weyl.** An important classical fact about matrix coefficients are the so-called Schur orthogonality relations. Roughly speaking, they describe how to compute the L^2-inner product of matrix coefficients. Since the L^2-inner product is closely related to convolution, this will enable us to easily compute convolutions of matrix coefficients.

Proposition 7.6. *(Schur orthogonality relations)*

(a) *If (π, V_π) and (σ, V_σ) are inequivalent irreducible representation of G, then*

$$\int_G \langle u_1, \pi(g)v_1\rangle \overline{\langle u_2, \sigma(g)v_2\rangle}\, dg = 0$$

for all $u_1, v_1 \in V_\pi$ and all $u_2, v_2 \in V_\sigma$.

(b) *If (π, V_π) is an irreducible unitary representation of G, then*

$$\int_G \overline{\langle u_1, \pi(g)v_1\rangle}\langle u_2, \pi(g)v_2\rangle\, dg = \frac{\overline{\langle u_1, u_2\rangle}\langle v_1, v_2\rangle}{d_\pi}$$

for all $u_1, v_1, u_2, v_2 \in V_\pi$.

Theorem 7.7. *(Peter-Weyl Theorem)*

(a) *The set*

$$\{\sqrt{d_\pi}\pi_{ij} : [\pi] \in \widehat{G}, 1 \leq i, j \leq d_\pi\}$$

is an orthonormal basis of $L^2(G)$.

(b) *Let (π, V_π) be a unitary representation of G. Then π can be written as a direct sum of irreducible unitary representations.*

As a consequence, we can write

$$L^2(G) \cong \widehat{\bigoplus_{[\pi]\in\hat{G}}} V_\pi \otimes V_\pi^*.$$

7.3. **Convolution of matrix coefficients.** In this subsection we will use Schur's orthogonality relations to compute convolutions of matrix coefficients. The information in this subsection is slightly less common but is relatively well known to experts.

Proposition 7.8. *Let π and σ be irreducible representation of G. Then*

$$\pi_{ij} * \pi_{kl} = \begin{cases} d_\pi^{-1}\pi_{il} & \text{if } j = k \\ 0 & \text{otherwise.} \end{cases}$$

If (π, V_π) and (σ, V_σ) are inequivalent representations of G, then $\pi_{ij} * \sigma_{kl} = 0$.

Proof. The second part follows from part (a) of Schur's orthogonality relations. For the first part we apply part (b):

$$\begin{aligned}
\pi_{ij} * \pi_{kl}(x) &= \int_G \pi_{ij}(g)\pi_{kl}(g^{-1}x)\, dg \\
&= \int_G \langle e_i, \pi(g)e_j\rangle\langle e_k, \pi(g^{-1}x)e_l\rangle\, dg \\
&= \int_G \langle e_i, \pi(g)e_j\rangle\langle \pi(g)e_k, \pi(x)e_l\rangle\, dg \\
&= \int_G \overline{\langle \pi(x)e_l, \pi(g)e_k\rangle}\langle e_i, \pi(g)e_j\rangle\, dg \\
&= d_\pi^{-1}\overline{\langle \pi(x)e_l, e_i\rangle}\langle e_k, e_j\rangle \\
&= d_\pi^{-1}\delta_{jk}\pi_{il}(x).
\end{aligned}$$

In the above, δ_{jk} is the usual Kronecker delta. □

8. Compact Lie groups

Let us now proceed to a particularly simple example of a double groupoid. In this case we consider a Lie group. A Lie group G which can be made into a double groupoid as follows:

$$\begin{array}{ccc} G & \rightrightarrows & G \\ \downarrow\downarrow & & \downarrow\downarrow \\ \{*\} & \rightrightarrows & \{*\} \end{array}$$

The groupoid structure on $G \rightrightarrows G$ is trivial with only identity arrows. Hence, the two "product" operations on G are given by the formulas:

$$g \circ g = g \qquad g \bullet h = gh$$

Where gh refers to the group operation.

Since the target fibers of $G \rightrightarrows G$ are trivial, one can just take the Haar system which declares the measure of a point to be one. In addition, let us choose a Haar measure μ on G with respect to the group product. Under these conventions, it is easy to see that $\overset{\circ}{*}$ is just the usual pointwise multiplication of functions on G and $\overset{\bullet}{*}$ is the usual convolution operation for the group:

$$u \overset{\circ}{*} v(g) = u(g)v(g) \qquad (u \overset{\bullet}{*} v)(g) = \int_{h \in G} u(h)v(h^{-1}g)d\mu$$

Even in this very simple case, the question of how these two product operations are related is not so trivial.

We now specialize the case where G is compact. This will be a stepping stone for studying the more complicated case of a noncommutative compact group. In the previous section we saw that a basis of functions for $C^\infty(G)$ can be obtained by taking the matrix coefficients of irreducible representations of G. Therefore, we already know how to compute $\overset{\circ}{*}$. From now on, to simplify the notation we will denote the pointwise product $\overset{\circ}{*}$ by Juxtaposition and the convolution $\overset{\bullet}{*}$ with just $*$.

8.1. **The Fourier groupoid.** By the Peter-Weyl theorem we know the set:

$$\mathcal{B} := \{\pi_{ij} \ : \ \pi \in \widehat{G}, 0 \leq i, j \leq d_\pi\}$$

generates a dense subalgebra of $C^\infty(G)$. Let $\widehat{G}$ denote the set of (equivalence classes) of irreducible representations of G. It is known that the subalgebra generated by this basis is actually the groupoid algebra of a discrete groupoid. In order to see this isomorphism let us define a very simple groupoid.

Definition 8.1. Let n be a natural number. The standard set $S(n)$ of n elements is the set of the first n positive integers. The standard pair groupoid on n-elements, $P(n) \rightrightarrows S(n)$ is the usual pair groupoid $P(n) = S(n) \times S(n) \rightrightarrows S(n)$ of $S(n)$.

Given a compact Lie group G we associate a groupoid formed by taking the disjoint union of standard pair groupoids. We will call this the "Fourier groupoid"

Definition 8.2. Let G be a compact Lie group. The *Fourier groupoid* of G is the discrete groupoid: $\mathcal{P}_G \rightrightarrows \mathcal{S}_G$ where:

$$\mathcal{P}_G := \bigsqcup_{\pi \in \widehat{G}} P(d_\pi) \qquad \mathcal{S}_G := \bigsqcup_{\pi \in \widehat{G}} S(d_\pi)$$

The groupoid structure on $\mathcal{P}_G$ is taken to be the disjoint union of the groupoid structures.

Our next theorem observes that the convolution algebra of G is isomorphic to the groupoid algebra of its Fourier groupoid.

Theorem 8.3. *The groupoid algebra of the Fourier groupoid $\mathcal{P}_G$ is isomorphic to a dense subalgebra of the convolution algebra $C^\infty(G)$.*

Proof. As usual, let $\mathcal{B}$ be a basis for $C^\infty(G)$ formed out of the matrix coefficients of irreducible representations.

Let

$$\widehat{\mathcal{B}} := \left\{ \frac{1}{d_\pi} \delta^\pi_{(i,j)} \in C(P(d_\pi)) \ : \ \pi \in \widehat{G},\ (i,j) \in P(d_\pi) \right\}$$

where $\delta^\pi_{(i,j)}$ denotes the usual delta indicator function for the element $(i,j) \in P(d_\pi)$. The set $\widehat{\mathcal{B}}$ constitutes a basis for functions on $\mathcal{P}_G$.

Now consider the bijection:

$$\mathcal{B} \to \widehat{\mathcal{B}} \qquad \pi_{ij} \mapsto \frac{1}{d_\pi} \delta^\pi_{(i,j)}$$

If we compute convolution of elements of $\mathcal{B}$ using Proposition 7.8 we see immediately that it behaves identically to the corresponding product on the groupoid algebra. Therefore, this bijection gives rise to an isomorphism between a dense subalgebra of the convolution algebra $C^\infty(G)$ and the discrete groupoid algebra. □

Remark 8.4. The above theorem is not novel, though our terminology might be. An easier proof exists, based on the 'endomorphism form' of the Peter-Weyl Theorem. In these versions of the Peter-Weyl theorem the convolution algebra is written as a direct sum of the $\mathrm{End}(V_\pi) \cong V_\pi \otimes V_\pi^*$. Algebras of endomorphisms are always isomorphic to pair groupoids so the theorem follows.

We remark that the isomorphism described in the above theorem is not canonical. It requires a choice of representative for each equivalence class in $\widehat{G}$ and a choice of basis for each such a representative.

8.2. **Convolution of non-irreducible matrix coefficients.** The previous section suggests to us that from the point of view of the convolution algebra, π_{ij} and σ_{ab} are "composable" if and only if $\pi = \sigma$ and $j = a$. Therefore, it is reasonable to expect that one might have a formula that expresses compatibility between convolution and pointwise multiplication as follows: For all π_{ij}, π_{jk}, σ_{ab} and σ_{bc} in $\mathcal{B}$:

$$(\pi_{ij} * \pi_{jk}) \cdot (\sigma_{ab} * \sigma_{bc}) = (\pi_{ij} \cdot \sigma_{ab}) * (\pi_{jk} \cdot \sigma_{bc}). \tag{8.1}$$

Unfortunately the above equation is false. For an example, we suggest the reader try calculating the case of SU(2) which is the simplest group we are aware of where this formula fails.

However, the main result of this section says that if we sum the above formula over j and b then it holds. In order to prove this we need to better understand convolution of matrix coefficients for *non-irreducible* representations. The reason we require non-irreducible representations is due to the following observation:

Let π and $\sigma \in \widehat{G}$ be irreducible representations of G of dimension n and m, respectively. Consider the pointwise product: $\pi_{ij}\sigma_{ab}$ for $1 \leq i, j \leq n$ and $1 \leq a, b \leq m$.

This pointwise product is a matrix coefficient for the representation $\pi \otimes \sigma$. Since we are assuming our representations come equipped with orthonormal bases, we can write this tensor product explicitly as the Kronecker product of matrices:

$$(\pi \otimes \sigma)_{(i\otimes a)(j\otimes b)} = \pi_{ij}\sigma_{ab}$$

where we recall from Example 3.3 that:

$$i \otimes a := (i-1)m + a \qquad j \otimes b := (j-1)m + b$$

However, the formula:

$$\pi_{ij} * \pi_{jk} = \frac{1}{d_\pi}\pi_{ik}$$

is not necessarily true for non-irreducible π. This means that once we start taking pointwise products, we can no longer use Proposition 7.8 to perform the calculation.

Our next proposition can be thought of as a weaker version of Proposition 7.8 which holds for arbitrary finite dimensional representations.

Proposition 8.5. *Let π be an n-dimensional representation of G. Let π_{ij} denote the matrix coefficients of π relative to an orthonormal basis. Then for all*

$$1 \leq i, k \leq n$$

we have that

$$\sum_{j=1}^{n} \pi_{ij} * \pi_{jk} = \pi_{jk}$$

Proof. Let us fix a basis for π and let us denote the associated matrix coefficients in the usual way. Note that by the formula for matrix multiplication, for any $g, h \in G$ we have that:

$$\pi_{ik}(gh) = \sum_{j=1}^{n} \pi(g)_{ij}\pi(h)_{ik}$$

Keeping this formula in mind we can prove the proposition by a direct calculation:

$$\begin{aligned}
\sum_{j=1}^{n} \pi_{ij} * \pi_{jk} &= \sum_{j=1}^{n} \int_G \pi(h)_{ij}\pi(h^{-1}g)_{jk}\, d\mu(h) \\
&= \int_G \sum_{j=1}^{n} \pi(h)_{ij}\pi(h^{-1}g)_{jk}\, d\mu(h) \\
&= \int_G \pi(g)_{ik}\, d\mu(h) \\
&= \pi(g)_{ik} \int_G 1\, d\mu(h) \\
&= \pi(g)
\end{aligned}$$

Where the last equality follows due to the fact that G is compact and the measure is normalized.

□

8.3. **Compatibility for matrix coefficients.** We can now prove a weaker form of Equation 8.1. This weaker form essentially says that the failure of Equation 8.1 vanishes "on average."

Theorem 8.6. *Let G be a compact Lie group and let $\mathcal{B}$ denote a basis formed from matrix coefficients. For all:*

$$0 \leq i, k \leq n, \quad 0 \leq a, c \leq m,$$

we have that

$$\sum_j \sum_b (\pi_{ij} * \pi_{jk}) \cdot (\sigma_{ab} * \sigma_{bc}) = \sum_j \sum_b (\pi_{ij} \cdot \sigma_{ab}) * (\pi_{jk} \cdot \sigma_{bc}). \tag{8.2}$$

Proof. Let $\alpha = \pi \otimes \sigma$ denote the Kronecker tensor product and m be the dimension of σ. Computing the left hand side of Equation 8.2 yields:

$$\begin{aligned}\sum_j \sum_b (\pi_{ij} * \pi_{jk}) \cdot (\sigma_{ab} * \sigma_{bc}) &= \pi_{ik} \cdot \sigma_{ac} \\ &= \alpha_{(i\otimes a)(k\otimes c)}.\end{aligned}$$

This is a straightforward application of Proposition 7.8.

On the other hand, computing the right hand side yields:

$$\begin{aligned}\sum_j \sum_b (\pi_{ij} \cdot \sigma_{ab}) * (\pi_{jk} \cdot \sigma_{bc}) &= \sum_j \sum_b \alpha_{(i\otimes a)(j\otimes b)} * \alpha_{(j\otimes b)(k\otimes c))} \\ &= \alpha_{(i\otimes a)(k\otimes c)}.\end{aligned}$$

where in the last equality we used Proposition 8.5. □

9. Compact singular Lie groups

We will now look at some examples of 2-groups where we can carry out a generalization of the computation we did for the noncommutative torus. We refer to these examples as "compact singular Lie groups" As singular spaces, they represent quotients of compact Lie groups by (possibly non-closed) discrete normal subgroups.

Throughout this section let G be a compact Lie group and let $K \subset G$ be a countable, normal subgroup. We do not assume that K is closed. An immediate consequence of K being countable and normal is that K is a subgroup of the center of G. There is a natural action of K on G via the inclusion and therefore we can form an action groupoid:

$$K \ltimes G \rightrightarrows G$$

As a reminder, the structure maps for this groupoid are as follows:

$$s(\kappa, g) = g \qquad t(\kappa, g) = \kappa g$$

$$(\kappa_1, \kappa_2 g) \circ (\kappa_2, g) = (\kappa_1 \kappa_2, g)$$

$$i(\kappa, g) = (\kappa^{-1}, \kappa g).$$

On the other hand, there is also the usual product group structure on $K \ltimes G$:

$$(\kappa_1, g_1) \bullet (\kappa_2, g_2) = (\kappa_1 \kappa_2, g_1 g_2).$$

These two composition operations come from a double groupoid of the form:

$$\begin{array}{ccc} K \ltimes G & \rightrightarrows & G \\ \downarrow\downarrow & & \downarrow\downarrow \\ \{*\} & \rightrightarrows & \{*\} \end{array}$$

Remark 9.1. The compatibility of these two operations relies on the fact that K is a subgroup of the center of G. If K is not countable, this construction still results in a Lie double groupoid so long as K is an immersed Lie subgroup of the center. However, we will focus on the countable case.

9.1. **Fourier groupoid for compact singular Lie groups.** When studying compact Lie groups, we saw that it was possible to construct a discrete groupoid whose groupoid algebra was isomorphic to the convolution algebra of the group. See Subsection 8.1. There is an analogous object for our noncommutative compact 2-groups.

Definition 9.2. Suppose G is compact and $K \subset G$ is a countable normal subgroup. The *Fourier groupoid* $\mathcal{P}_{\mathcal{G}}$ of the double groupoid $\mathcal{G} := K \ltimes G$ is just the product groupoid $K \times \mathcal{P}_G \rightrightarrows \mathcal{S}_G$.

Remark 9.3. The role of the Fourier groupoid of $K \ltimes G$ is similar to the one for G. Note that this version of the Fourier groupoid is taking into account the "double groupoid"-ness of $\mathcal{G}$. Our previous definition for Fourier groupoid cannot be applied to $K \ltimes G$ unless K is finite and so $K \ltimes G$ is compact. However, the Fourier groupoid of $K \times G$ as a compact group is not quite the same as the Fourier groupoid of $K \ltimes G$ thought of as a double groupoid. This is because the double groupoid version seems to "ignore" the K-component.

It would be nice to have a more intrinsic definition of the Fourier groupoid that could be applied to a more general class of Lie groupoids. However, at this time it is not clear what the correct definition should be.

Theorem 9.4. *Let $\mathcal{G} = K \ltimes G$ be a compact singular Lie group. The groupoid algebra of $\mathcal{P}_{\mathcal{G}}$ is isomorphic to a dense subalgebra of $\left(C_c^\infty(\mathcal{G}), \overset{\bullet}{*}\right)$.*

9.2. **Basis of matrix coefficients.** We would like to construct a convenient basis for $C_c^\infty(K \ltimes G)$. Since we have assumed that G is compact we already have a good basis for $C_c^\infty(G)$ constructed from the matrix coefficients. Since $K \ltimes G$ is just a product space, we can construct a basis for its space of functions.

We will continue to use the notational conventions for matrix coefficients that we established in earlier sections. As a reminder: Throughout this section we write $\widehat{G}$ to denote the set of equivalence classes of irreducible representations of G. We assume that we have chosen a representative from each equivalence class that is equipped with an orthonormal basis. For $\pi \in \widehat{G}$ we write $\pi_{ij} \in C^\infty(G)$ to denote i,j-th matrix coefficient. We write d_π to denote the dimension of the underlying vector space V_π of the representation.

Definition 9.5. Let π be an arbitrary unitary representation of G and suppose π_{ij} is a matrix coefficient for $1 \leq i,j \leq d_\pi$. Given $\kappa \in K$ let $\pi_{ij}^\kappa \in C_c^\infty(K \ltimes G)$ be defined as follows:

$$\pi^\kappa(\lambda, g) = \begin{cases} \pi_{ij}(g) & \text{if } \kappa = \lambda \\ 0 & \text{otherwise.} \end{cases}$$

The basis of matrix coefficients is the set:

$$\mathcal{B} := \{\pi_{ij}^{\kappa} \ : \ \pi \in \widehat{G}, \kappa \in K, \ 0 \leq i, j \leq d_\pi\}.$$

Note that we have defined π_{ij}^{κ} for all representations but only the irreducible representations are featured in this basis.

It follows from the Peter-Weyl theorem that the span of $\mathcal{B}$ in $C_c^\infty(K \ltimes G)$ is a dense subspace.

9.3. **Computing convolution products.** We can now compute the two convolutions on $C_c^\infty(K \ltimes G)$ in terms of our basis $\mathcal{B}$. Our first formula is a generalization of Proposition 7.8 which takes into account the "K"-components.

Proposition 9.6. *Let π and σ be irreducible representations of G. Consider two elements π_{ij}^{κ} and σ_{ab}^{λ} in our basis $\mathcal{B}$. Then we have that:*

$$\pi_{ij}^{\kappa} \overset{\bullet}{*} \sigma_{ab}^{\lambda} = \begin{cases} d_\pi^{-1} \pi_{ib}^{\kappa\lambda} & \text{if } \pi \simeq \sigma, j = a \\ 0 & \text{otherwise.} \end{cases}$$

Proof. The basis $\mathcal{B}$ can be thought of as the product of the matrix coefficient basis on G with the delta function basis for $C_c^\infty(K)$. Since we are using the product group structure and product Haar measures, the convolution product is obtained by performing convolution in each component separately.

More explicitly, we can check this formula by a direct calculation:

$$\begin{aligned}
\pi_{ij}^{\kappa} \overset{\bullet}{*} \sigma_{ab}^{\lambda}(x, y) &= \int_{K \times G} \pi_{ij}^{\kappa}(\ell, g) \sigma_{ab}^{\lambda}(\ell^{-1}x, g^{-1}y) \, d\mu^H(\ell, g) \\
&= \sum_{\ell} \int_G \pi_{ij}^{\kappa}(\ell, g) \sigma_{ab}^{\lambda}(\ell^{-1}x, g^{-1}y) \, dg \\
&= \int_G \pi_{ij}(g) \sigma_{ab}^{\lambda}(\kappa^{-1}x, g^{-1}y) \, dg \\
&= \begin{cases} \int_G \pi_{ij}(g) \sigma_{ab}(g^{-1}y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \pi_{ij} * \sigma_{ab}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} d_\pi^{-1} \pi_{ib}(y) & \text{if } \kappa\lambda = x, \pi \simeq \sigma, j = a \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} d_\pi^{-1} \pi_{ib}^{\lambda\gamma}(x, y) & \text{if } \pi \simeq \sigma, j = a \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

□

Now let us compute the other convolution product. This one is more complex. However, the computation is familiar to those who have seen convolution algebras constructed out of group actions.

Given a function $f \in C_c^\infty(K \ltimes G)$, there is a natural action of K by left translation. Given $\kappa \in K$ let:

$$L_\kappa f \colon G \to \mathbb{C} \qquad (L_\kappa f)(x) := f(\kappa x)$$

Our next lemma

Lemma 9.7. *Let $\pi \in \widehat{G}$ be an irreducible representation. There exists a group homomorphism:*

$$\xi_\pi \colon K \to \mathbb{C}^\times$$

with the property that for all $\kappa \in K$ and matrix coefficients π_{ij} we have that $L_\kappa(\pi_{ij}) = \xi_\pi(\kappa)\pi_{ij}$.

Proof. Recall that $K \subset Z(G)$ is a subgroup of the center. Furthermore, if $\gamma \in Z(G)$, then $\pi(\gamma)$ commutes with $\pi(g)$ for all $g \in G$. Since π is irreducible unitary, by Schur's lemma, $\pi(\gamma)$ is scalar. Therefore, we have a group homomorphism:

$$\Phi \colon Z(G) \to \mathbb{C}^\times$$

Let ξ be restriction of Φ to K.

From the definition of ξ it follows that:

$$\pi(\kappa x) = \pi(\kappa)\pi(x) = \xi(\kappa)\pi(x)$$

Then for any matrix coefficient π_{ij} it follows that:

$$L_\kappa \pi_{ij}(x) = \pi(\kappa x)_{ij} = \xi(\kappa)\pi_{ij}(x).$$

□

Definition 9.8. Suppose that π is an irreducible unitary representation of G. If K is a subgroup of the center of G then we say that the *K-character on π* is the homomorphism $\xi_\pi \colon K \to \mathbb{C}^\times$ from Lemma 9.7.

By utilizing the K-characters of the irreducible representations, we can now compute the $\overset{\circ}{*}$ convolution product.

Proposition 9.9. *Let $\pi, \sigma \in \widehat{G}$. Then for all $\kappa, \lambda \in K$ and matrix coefficients π_{ij}, σ_{ab}, we have that:*

$$\pi^\kappa_{ij} \overset{\circ}{*} \sigma^\lambda_{\kappa,ab} = \xi_\pi(\lambda)(\pi \otimes \sigma)^{\kappa\lambda}_{(i\otimes a)(j\otimes b)}.$$

Note that the result of this product is not an element of our basis $\mathcal{B}$ unless $\pi \otimes \sigma$ is irreducible. However, it can be written uniquely in terms of elements of $\mathcal{B}$. As we remarked earlier in Subsection 8.3, this computation is rather non-trivial.

Proof. The proof is by a direct calculation and utilizing the definition of ξ_π.

$$\begin{aligned}
\pi_{ij}^{\kappa} \overset{\circ}{*} \sigma_{ab}^{\lambda}(x,y) &= \sum_{\ell \in K} \pi_{ij}^{\kappa}(\ell, \ell^{-1}xy)\sigma_{ab}^{\lambda}(\ell^{-1}x, y) \\
&= \pi_{ij}(\kappa^{-1}xy)\sigma_{ab}^{\lambda}(\kappa^{-1}x, y) \\
&= \begin{cases} \pi_{ij}(\kappa^{-1}xy)\sigma_{ab}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \pi_{ij}(\lambda y)\sigma_{ab}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} L_\lambda \pi_{ij}(y)\sigma_{ab}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \xi_\pi(\lambda)\pi_{ij}(y)\sigma_{ab}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise.} \end{cases}
\end{aligned}$$

□

Example 9.10. In the case of the noncommutative torus the action of $\mathbb{Z}$ by an irrational rotation $r \notin \pi\mathbb{Q}$ is given by an injective group homomorphism:

$$\mathbb{Z} \hookrightarrow \mathbb{S}^1 \qquad k \mapsto e^{ikr}$$

The group K is the image of this homomorphism. The irreducible representations are all one dimensional and are indexed by integers which represent the frequency of the representation. Given an irreducible representation π_n with frequency n then the K-character of π_n is just the homomorphism:

$$\xi_n \colon K \to \mathbb{C}^\times \qquad \xi_n(\kappa) = \kappa^n$$

Note that if $\kappa = e^{ikr} \in \mathbb{S}^1$ then $\kappa^n = e^{irkn}$ which is the same factor that appeared in the original calculation of $\overset{\circ}{*}$ in Proposition 6.1 (take $k = c, n = b$).

9.4. **Compatibility for matrix coefficients.** We will now prove a version of Proposition 8.5 and Theorem 8.6 that is generalized to the case of a compact singular Lie group.

Proposition 9.11. *Suppose π is a finite dimensional representation of G and let κ, λ be elements of K. Then we have that:*

$$\sum_j \pi_{ij}^{\kappa} \overset{\bullet}{*} \pi_{jk}^{\lambda} = \pi_{ik}^{\kappa\lambda}. \tag{9.1}$$

Proof. This is essentially a corollary of Proposition 8.5.

We prove prove it by computing the right hand side at a point $(x, y) \in K \ltimes G$:

$$\begin{aligned}
\sum_j \pi_{ij}^{\kappa} \overset{\bullet}{*} \pi_{jk}^{\lambda}(x,y) &= \sum_j \int_{K\times G} \pi_{ij}^{\kappa}(\ell, g)\pi_{jk}^{\lambda}(\ell^{-1}x, g^{-1}y)\, d\mu^H(\ell, g) \\
&= \sum_j \sum_{\ell} \int_G \pi_{ij}^{\kappa}(\ell, g)\pi_{jk}^{\lambda}(\ell^{-1}x, g^{-1}y)\, dg \\
&= \sum_j \int_G \pi_{ij}(g)\pi_{jk}^{\lambda}(\kappa^{-1}x, g^{-1}y)\, dg \\
&= \begin{cases} \sum_j \int_G \pi_{ij}(g)\pi_{jk}(g^{-1}y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \sum_j \pi_{ij} * \pi_{jk}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \begin{cases} \pi_{ik}(y) & \text{if } \kappa\lambda = x \\ 0 & \text{otherwise} \end{cases} \\
&= \pi_{ik}^{\kappa\lambda}(x,y)
\end{aligned}$$

where the second to last equality is obtained by applying Proposition 8.5. □

Note that in the above proposition we have not assumed that π is irreducible so we cannot directly apply the calculation from Proposition 9.6. With this proposition in hand, we can now prove our last theorem.

Theorem 9.12. *The set*

$$\mathcal{B} := \{\pi_{ij}^{\kappa} \ : \ \pi \in \widehat{G}, \kappa \in K, 0 \leq i, j \leq \dim(\pi)\}$$

generates a dense subspace of $C_c^{\infty}(K \ltimes G, \mathbb{C})$. *Furthermore, for all:*

$$\kappa_1, \kappa_2, \lambda_1, \lambda_2 \in K, \quad 0 \leq i, k \leq n, \quad 0 \leq a, c \leq m$$

we have that

$$\sum_j \sum_b (\pi_{ij}^{\kappa_1} \overset{\bullet}{*} \pi_{jk}^{\kappa_2}) \overset{\circ}{*} (\sigma_{ab}^{\lambda_1} \overset{\bullet}{*} \sigma_{bc}^{\lambda_2}) = \sum_j \sum_b (\pi_{ij}^{\kappa_1} \overset{\circ}{*} \sigma_{ab}^{\lambda_1}) \overset{\bullet}{*} (\pi_{jk}^{\kappa_2} \overset{\circ}{*} \sigma_{bc}^{\lambda_2}). \tag{9.2}$$

Proof. The first part of the theorem is something we have already observed. It is a straightforward corollary of the Peter-Weyl theorem.

We will focus on proving Equation 9.2. The calculation is essentially the same as the one in Theorem 8.6.

Let $\alpha = \pi \otimes \sigma$ denote the Kronecker tensor product and m be the dimension of σ. Let ξ_π be the K-character of π. Computing the left hand side of Equation 9.2 yields:

$$\begin{aligned}
\sum_j \sum_b (\pi_{ij}^{\kappa_1} \overset{\bullet}{*} \pi_{jk}^{\kappa_2}) \overset{\circ}{*} (\sigma_{ab}^{\lambda_1} \overset{\bullet}{*} \sigma_{bc}^{\lambda_2}) &= \pi_{ik}^{\kappa_1\kappa_2} \overset{\circ}{*} \sigma_{ac}^{\lambda_1\lambda_2} \\
&= \xi_\pi(\lambda_1\lambda_2)\alpha_{(i\otimes a)(k\otimes c)}^{\kappa_1\kappa_2\lambda_1\lambda_2}.
\end{aligned}$$

In the first equality we used our computation of $\overset{\bullet}{*}$ from Proposition 9.6 and in the second equality we applied our computation for $\overset{\circ}{*}$ from Proposition 9.9.

On the other hand, computing the right hand side yields:

$$\begin{aligned}\sum_j \sum_b (\pi_{ij}^{\kappa_1} \overset{\circ}{*} \sigma_{ab}^{\lambda_1}) \overset{\bullet}{*} (\pi_{jk}^{\kappa_2} \overset{\circ}{*} \sigma_{bc}^{\lambda_2}) &= \sum_j \sum_b \xi_\pi(\lambda_1)\alpha^{\kappa_1\lambda_1}_{(i\otimes a)(j\otimes b)} \overset{\bullet}{*} \xi_\pi(\lambda_2)\alpha^{\kappa_1\lambda_2}_{(j\otimes b)(k\otimes c)} \\ &= \xi_\pi(\lambda_1)\xi_\pi(\lambda_2) \sum_j \sum_b \alpha^{\kappa_1\lambda_1}_{(i\otimes a)(j\otimes b)} \overset{\bullet}{*} \alpha^{\kappa_1\lambda_2}_{(j\otimes b)(k\otimes c)} \\ &= \xi_\pi(\lambda_1)(\lambda_2)\alpha^{\kappa_1\lambda_1\kappa_2\lambda_2}_{(i\otimes a)(k\otimes c)}.\end{aligned}$$

where in the last equality we used Proposition 9.11. Since ξ is a homomorphism and K is abelian, the results of both of these computations are equal. □

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